

Wanglong Lu

Senior Data Scientist | Applied Multimodal Intelligence

St. John's, NL, Canada | lwlxhl@gmail.com

Website GitHub LinkedIn Google Scholar

Profile

Applied ML researcher and Senior Data Scientist at Nasdaq, with a Ph.D. in Computer Science. Works on financial ML, multimodal image editing, and efficient adaptation, with experience deploying ML workflows on AWS.

First-author **IEEE TNNLS** and **TVCG** publications; **NeurIPS** research integrated into **Hugging Face PEFT**.

Technical Skills

Core: Python, PyTorch, SQL, Git, Bash, C++, CUDA; experimental design and feature engineering.

ML & GenAI: Latent diffusion, Transformers, multimodal AI, PEFT/LoRA, Hugging Face, RAG, agentic tools.

ML systems: FastAPI, Mangum, PostgreSQL; AWS SageMaker, Bedrock, Lambda, S3, Docker, SLURM.

Professional Experience

Senior Data Scientist, AI/Analytics – Nasdaq

Mar. 2025 – Present

St. John's, NL, Canada

- Designed cross-scale ensemble feature selection for **cheque-fraud ML (CFML)**, removing **77.2% of input features** while retaining competitive performance in evaluations.
- Deployed and integrated an **existing UI/backend prototype** for a cross-team ML training platform into an internal workspace; configured networking and IAM access, added **PostgreSQL-backed user-state persistence**, and deployed **FastAPI** services on **AWS Lambda using Mangum**.
- Built a Python job-submission workflow that packages local scripts and dependencies for **SageMaker Processing, Training, and Hyperparameter Tuning**; linked jobs with **SageMaker Pipelines**.
- Developed a **Bedrock-based feature-formula validator** with structured logical-consistency checks and reliability scores to flag hard-coded errors in feature-engineering formulas.
- Built a **RAG-based incident-response assistant** as an internal hackathon project, retrieving troubleshooting context for on-call alerts through the company's GenAI platform.

AI Algorithm Intern – Nasdaq Verafin

May 2024 – Jan. 2025

St. John's, NL, Canada

- Designed and benchmarked diffusion-based restoration and inpainting for degraded cheque images using Python and PyTorch, comparing with existing restoration methods.
- Developed patch-level Gaussian latent modeling for image verification; evaluated the approach and presented results to cross-functional colleagues.

AI Intern – Beaufort Solutions

Jul. 2022 – Nov. 2022

St. John's, NL, Canada

- Designed a data-collection pipeline and few-shot vision-language classifier for photo-book curation; led data collection and evaluation, contributing to TEG (Journal of Electronic Imaging, 2024).

AI Algorithm Intern – Zhongkong Hanlian

Dec. 2017 – Aug. 2018

Hangzhou, China

- Built vehicle-logo recognition models and annotation tools in Python, C++, and Java; supported real-time deployment at community security gates.

Selected Research Publications

UltraDiffEdit – High-Resolution Generative Editing First author; IEEE TNNLS 2026

- Developed tuning-free editing with multi-patch encoding, global-local denoising, and progressive sampling; demonstrated **8K editing on a single RTX 3090**. Code; video.

FACEMUG – Multimodal Local Facial Editing First author; IEEE TVCG 2024

- Designed multimodal feature fusion and self-supervised latent warping for local facial edits across **six input modalities**, including text, sketches, and exemplars, while preserving unedited regions.

AdaMSS – Parameter-Efficient Adaptation Co-author; NeurIPS 2025

- Contributed to multi-subspace fine-tuning; reported ViT-Large results exceeded LoRA’s average accuracy across seven tasks using **15.4% of LoRA’s trainable parameters**. Method **integrated into Hugging Face PEFT**. Code; PEFT integration.

DocPure – Unified Document Restoration Co-first author; IEEE TCSVT 2026

- Co-authored prompt-free document restoration using degradation-aware structure extraction and structure-guided wavelet modulation across multiple degradations. Code.

GRIG – Data-Efficient Image Inpainting First author; CVM 2025

- Developed iterative residual inpainting using CNN features, Transformer reasoning, and forgery-patch adversarial training; evaluated on **10 few-shot datasets** using FID and LPIPS. Code; benchmarks.

Visual Style Prompt Learning First author; Pattern Recognition 2025

- Developed diffusion-generated visual prompts and style-modulated aggregation for blind face restoration. Code.

Full publication record: [Google Scholar](#) and [publication portfolio](#).

AI Engineering Prototypes & Deployment

Local Agentic Code Editor Nasdaq internal prototype; 2026

- Prototyped a local AI code editor connecting **language-model APIs with local tool execution**; an **agentic loop** selects code writing, code review, or shell-tool execution to support code generation and debugging.

Education

Ph.D., Computer Science – Memorial University of Newfoundland, Canada 2021–2025

M.Sc., Computer Software and Theory – Wenzhou University, China 2018–2021

B.Sc., Digital Media Technology – Communication University of Zhejiang, China 2014–2018

Selected Recognition & Service

Outstanding Research Award (Ph.D.), Memorial University, 2023; four granted Chinese invention patents.

Mentored **10 students** in generative vision and applied ML; supported industry research on cheque OCR and tabular-data generation.

Invited reviewer for **IEEE TPAMI, TIP, TMM, TCSVT**, Pattern Recognition, Knowledge-Based Systems, Neurocomputing, Applied Soft Computing, and ECCV 2026.

Presented feature-selection research to **60+ attendees** (2026); invited speaker, IEEE Newfoundland and Labrador Section AGM (2026); guest lecturer on deep generative models, Memorial University (2025).